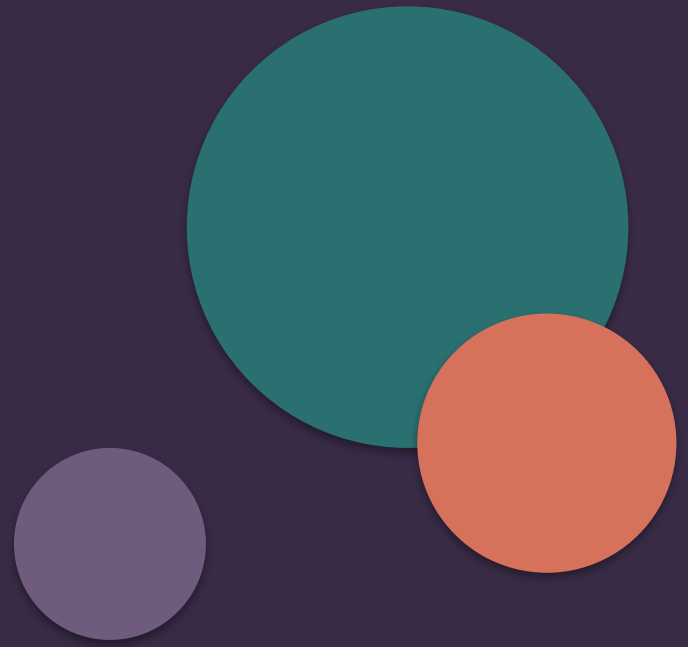
DDCA • CO6

8-bit Signed Binary Representation



From Boolean logic and gates to an 8-bit signed value in a processor

Logic gates • Boolean algebra • CPU extensions

Bits, patterns and signed meaning

A bit is 0 or 1. Eight bits form a pattern; two’s complement defines how the pattern represents a signed integer.

8-BIT EXAMPLE

```
1111 1011
```

Interpretation

In 8-bit two’s complement, 1111 1011 represents −5.

Bit values

Each bit is a logic level: 0 or 1.

Signed range

−128 to +127

Two’s-complement weights

For an 8-bit two’s-complement value, the leftmost bit has weight −128; the other positions have positive powers of two.

Bit weights

−128 64 32 16 8 4 2 1

PATTERN

1111 1011

Add the active weights

−128 + 64 + 32 + 16 + 8 + 0 + 2 + 1 = −5

From truth table to gate circuit

Boolean expressions describe how outputs depend on input bits.

Example: XOR

A XOR B = 1 when A and B are different.

00 → 0 01 → 1

10 → 1 11 → 0

Sum-of-products (SOP)

Write the output as OR of AND terms for rows where the output is 1.

For XOR: A′B + AB′

SOP is one way to translate a truth table into a logic circuit, as covered in CO6.

Why NAND and NOR are universal

NAND-only or NOR-only circuits can implement every Boolean function.

Build NOT from NAND

Connect both NAND inputs to A:

A NAND A = NOT A

Build AND from NAND

First: X = A NAND B

Then: Y = X NAND X = A AND B

This matters because complex logic can be reduced to a single universal gate type.

An 8-bit register: gates to stored data

The binary value is stored by sequential elements, while combinational logic can transform it.

1. Input bits

An 8-bit input carries eight 0/1 values.

2. Eight flip-flops

Each flip-flop stores one bit, usually updating on a clock edge.

3. Interpret value

Two’s complement maps the stored pattern to a signed integer.

Example stored pattern 1111 1011 is interpreted as −5 when the system uses 8-bit two’s complement.

Using signed values in a RISC-V CPU

The CPU’s ALU processes bit patterns; control logic selects which operation the ALU performs.

Operand

An 8-bit example value can be interpreted as signed two’s complement.

ALU

ADD, SUB, AND, OR and SLT are among the operations named in the syllabus.

Control unit

Decodes instruction fields and sets control signals to steer datapath behaviour.

The full course uses a 32-bit RISC-V datapath; 8-bit signed examples here are simplified demonstrations of bit-level ideas.

Extending the machine

CO6 asks students to extend the CPU after building its layers.

New instruction

Example: add XOR as an instruction by updating the ALU and control unit.

Memory-mapped output

Use a CPU store to control an output peripheral such as LEDs or a display.

Performance

Clock period, CPI, pipelining, caching and parallelism shape performance.

Quick practice

Check your understanding.

- 1. What is the 8-bit signed range?
- 2. What decimal value is 1111 1011?
- 3. Why are NAND and NOR universal?
- 4. What does SOP stand for?

Answers

1. −128 to +127
2. −5
3. Any Boolean function can be built from either gate type
4. Sum of Products

Key takeaways

The bit pattern is the foundation; Boolean logic and architecture make it useful.

- Eight bits form a pattern; two’s complement gives it signed meaning.
- For 8 bits, the range is −128 to +127.
- Truth tables can be expressed with Boolean equations and SOP.
- NAND and NOR are universal gates.
- Registers store patterns; the ALU computes; the control unit selects operations.
- CO6 extends the CPU with instructions and memory-mapped peripherals.

Thank You

Logic Gates & Architecture | 8-bit Signed Binary Representation